COURSE SYLLABUS Spring 2026

IST 387/600: Introduction to Applied Data Science

Lecture Instructor: Dr. Preeti Jagadev (pjagadev@syr.edu)

Lecture Meetings: Thursdays, 11.00 AM-12:20 PM, Maxwell Hall Aud

Office Hours: Upon request

Lab Meeting	Lab Instructor	Email
IST387M002: Tu 11.00 AM – 12.20 PM, Hinds 243A	Dr. Preeti Jagadev	pjagadev@syr.edu
IST387M003: Tu 11:00AM - 12:20PM, Hinds Hall 018	Una Joh	sjoh01@syr.edu
IST387M004: Tu 9:30 AM – 10:50 AM, Hinds Hall 117	Una Joh	sjoh01@syr.edu
IST387M006: Tu 11.00 AM – 12.20 PM, LSB 300	Bo Zhang	bzhang49@syr.edu
IST600M002: Th 12:30 PM – 1:50 PM, Hinds Hall 011	Bo Zhang	bzhang49@syr.edu

About Myself (Your course Instructor!):

Welcome to **IST 387/600: Introduction to Applied Data Science**, where we turn messy data into meaningful insights, and occasionally stare at a screen wondering why the numbers refuse to cooperate (ha!). The deadlines are real, and "it worked yesterday" is not an explanation.

I'm your instructor, Dr. Preeti Jagadev, all the way from Goa, India. I teach this course because I absolutely love the fascinating world of Data Science, and I can't wait to share that excitement with you. When I'm not at the university decoding human and machine language, you can usually find me spending time with my baby, or experimenting in the kitchen, or baking things that may or may not survive long enough to be photographed.

I want us to have a great semester. If you **read the notes**, **respect the deadlines**, and **run the code** instead of just staring at it and hoping for osmosis, you're going to love my class. However, if you treat the deadlines like 'suggestions' and the notes like 'terms and conditions' that you just scroll past, well, you probably won't like my class very much at all.

I'm here to make sure you have the tools, support, and encouragement to succeed. By the end of the semester, my hope is that you walk away not just with new skills, but with the confidence to tackle any challenge, in tech, or in life. Let's make this a semester you'll be proud of.

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Important Notes:

- You must own a functioning laptop with **R** and **RStudio** installed or create an account on ***posit.cloud*** (formerly *rstudio.cloud*).
- You must bring your laptop to every class meeting, both lecture and lab.
- Emails addressed to the professor should have the subject line "**IST 387: XYZ**" where XYZ summarizes your question.

Course Description:

This course provides a comprehensive introduction to the lifecycle of data, focusing on the essential standards, technologies, and methodologies used to organize, manage, and preserve information. Further, we will examine the critical pillars of modern data science: ethics, quality control, and the responsible publication of findings.

Throughout the semester, you will transition from theoretical understanding to practical application in the following areas:

- Master the end-to-end process of data collection, transformation, and curation.
- Explore core concepts in applied statistics, machine learning, and text mining.
- Learn to translate complex datasets into clear, actionable visual narratives.
- Discuss the social and professional responsibilities of handling sensitive information and ensuring data integrity.

The technical backbone of this course is R, one of the most powerful tools in the global data science community. By developing proficiency in R, you are not just completing coursework, you are acquiring a highly marketable skill set that is in high demand across tech, research, and corporate sectors.

Course Structure:

The course will consist of one **large Lecture** and one **lab section** each week. Each is 80 minutes. The large lecture will include both undergraduate and graduate students. During the large lectures, we will explore key concepts and examine the use of those concepts within R. Prior to attending the lectures, it is expected that you've completed the reading for that week.

Your lab section will only be undergraduate students (i.e., students taking 387) or all graduate students (i.e., students taking 600). The weekly lab sessions will review key concepts: let you practice R coding and will also review any student questions that have arisen during the week. During the lab and the homework, you will have a chance to practice and apply your knowledge. There will also be an **exam** that will include questions about the lecture material as well as R-coding questions. There will also be a final **project** which you will complete during the semester. This project will allow you to apply what you have learnt within the class to a real-world data problem, where your task is to understand the domain and the data available to determine how to best provide insight and wisdom from all the data that might be available.

Audience: Undergraduate and graduate students.

Credits: 3 credit hours.

Learning Objectives:

After taking this course, the students will be expected to understand:

- Data characteristics, ethics, and integrity.
- Scripting and data management in R/RStudio.
- Screen, clean, and transform complex datasets.
- Organize data throughout the entire project lifecycle.

- Match specific problems with appropriate analytical techniques.
- Translate technical findings into clear visualizations and actionable insights for decision-makers.

Text Recommended (Optional):
"Data Science for Business with R (2021) by Jeffrey S. Saltz and Jeffrey M. Stanton" (ISBN: 978-1544370453).

The book is available in paperback and electronic versions on Amazon: https://www.amazon.com/Data-Science-Business-Jeffrey-Saltz/dp/1544370458.

Grading: The work for this class will involve the following:

- **Homework (24%):** The homework assignments (each worth 2 points) are designed for you to practice the necessary skills in carrying out data processing, analysis, and management tasks. The code you create for the homework assignments will be instrumental in helping you to complete the exam successfully.
- **Weekly (quiz-based) Challenges (12%):** You will have an opportunity to participate in class challenges in the form of 4-6 multiple choice question quizzes. You need to complete these quizzes during each class meeting.
- **Lab Assignments (12%):** While you are in lab, you will work on an in-class assignment. At the end of each lab session, you will submit a copy of your work (R code) on Blackboard.
- **Exam (24%):** The exam is designed to take 80 minutes to complete. It will be open-book and open-notes, but you may not discuss the exam with other students.
- **Final Project (25%):** This is a predictive analytics effort that will be completed during the last half of the course. More information will be made available during the course.
- **Participation (3%):** Includes professionalism in class and attending class.

Each assigned work will be graded on the scale as specified for the component. All individual grades are summed at the end of the semester, and the final letter grade is determined using the following scale.

Grade	Total Points	Expectation
A	100 - 93	Your work is outstanding, always complete, and submitted on time.
A-	92 - 90	
B+	89 - 87	Your work is what we expect from a serious, professional student, but may contain occasional errors, omissions, or inconsistencies.
B	86 - 83	
B-	82 - 80	
C+	79 - 77	Your work falls below what we expect but is adequate for completing the course.
C	76 - 73	
C-	72 - 70	
D	69 - 60	Your work falls notably below what we expect and contains one or more deficiencies.
F	59 - 0	Your work is insufficient to pass the course.

Course Policies:

As a Syracuse University student, the faculty expects you to uphold the highest standards of professionality including honesty, integrity, respectful communications, lifelong commitment to learning, supporting others, punctuality, and appreciation of the diverse talents and capabilities of colleagues and coworkers.

Grades will be based on the quality and originality of your work, so bring your best ideas and curiosity. My judgment on submitted work is final. There are no extra credit assignments, and late submissions lose points fast:

1 day late → 20% off,
2 days → 40% off,
3 days → 60% off,
4 days → 80% off,
5+ days → zero (ouch!).

The **least two lab grades** for all the students will be dropped at the end of the semester.

Final grades won't be rounded, so every point counts!

Respectful communication with your peers and the instructor is expected; **the latter includes the avoidance of "grade negotiations".**

Originality is paramount. You may not share code with other students or borrow code from other students (please read our University's Academic Integrity Statement). Any work that was produced by others (such as code fragments that appear online) must have an accompanying source citation. Similarly, if you use AI to generate a chunk of code, you must make note of that. If you do not do this, it is an academic integrity violation. When in doubt, ASK! If the code did not originate from you, you must clearly disclose where it came from.

On-time arrival is expected. Late arrival or early departures from class that are unexcused will be considered in calculating attendance and participation. **Late by more than 5 minutes?** Time clearly moved faster for you! Your participation grade slows down by 10%.

Students who wish to dispute a grade may resubmit the assignment for regrading with a one-page statement of explanation of why the paper should be regraded. If the student resubmits, the assignment will be regraded, which means the grade may go up, down, or stay the same. Except for extraordinary circumstances, no appeal for an individual assignment or project will be considered later than two weeks after the assignment was graded.

Excused absences include, but are not limited to, as Lab Instructors may excuse absences on a case-by-case basis:

- Absences, such as hospitalization or a death in the family, that are verified through an Absence Notification email to the Lab Instructor.
- An athletic obligation with an official SU athletic team; and
- Absences through religious observances will also be excused if you have registered this information on MySlice.

The use of electronic devices other than your laptop is not required. Accommodation and modifications to utilize electronic devices should be discussed with your Lab Instructor. Smart phones need to be put on silent mode.

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Course Schedule (subject to change)**:**

Week/ Date of Lecture	Lecture Topic	Required Readings	Focus of Lab	Assignments*
Lab 0: 13th Jan (IST 387) and 15th Jan (IST 600)	-	-	Lab 0	Install R & RStudio
Lecture 1/ Jan 15	Data Science Overview	Ch 1: Begin at the Beginning With R	Using R; Working with vectors	HW 1 Assigned
Lecture 2/ Jan 22	Using R for data processing and handling	Ch 2: Rows & Columns Ch 3: Data Munging	Working with data frames	HW 2 Assigned
Lecture 3/ Jan 29	Descriptive Statistics & Functions	Ch 4: What's my Function? Ch 5: Beer, Farms & Peas	Descriptive statistics & writing functions	HW 3 Assigned
Lecture 4/ Feb 5	Sampling & Inferential Statistics	Ch 6: Sample in a Jar	Sampling	HW 4 Assigned
Lecture 5/ Feb 12	Connecting with External Data Sources	Ch 7: Accessing Data (storage wars)	Using JSON	HW 5 Assigned
Lecture 6/ Feb 19	Introduction to Visualization	Ch 8: Pictures vs numbers	Using ggplot2	HW 6 Assigned
Lecture 7/ Feb 26	Working with Map Data	Ch 9: Map Mash-Up	Maps	HW 7 Assigned
Lecture 8/ March 5	Linear Modeling (LM)	Ch 10: Lining Up Our Models	Working with LM **No Lab Next Week (Spring Break)**	HW 8 Assigned
March 12			**No Lecture and Lab (Spring Break) Classes Resume from 16th March**	
Lecture 9/March 19	Supervised Learning	Ch 11: What's your vector, Victor?	Support Vector Machines and Trees	HW 9 Assigned
Lecture 10/ March 26	Unsupervised Learning & Ethical Challenges	Ch 12: Hi Ho, Hi Ho Data Mining We Go	Association Rules Mining	HW 10 Assigned
April 2	**No Lecture, Study for Exam**	-	**Exam in lab Next Week**	-
Lecture 11/April 9	Interactive Apps	Ch 14: Shiny Web Apps	Shiny	HW 11 Assigned
Lecture 12/April 16	Text Mining & ChatGPT	Ch 13: Word perfect	Word clouds and sentiment analysis	HW 12 Assigned
April 23	**No Lecture – Group Project Time**		**Final Project**	-

* You must submit all assignments to Blackboard prior to the deadline.

SYRACUSE UNIVERSITY STUDENT POLICIES & SERVICES

Syllabi Appendix for the iSchool - Effective August 2024

iSchool Values

Excellence; Discovery & Innovation; Integrity;
Diversity & Inclusion; Global Citizenship and Engagement

Syracuse University has a variety of policies designed to guarantee that students live and study in a community respectful of their needs and those of fellow students. **These statements are an official part of this course syllabus**.

University Attendance Policy

Attendance in classes is expected in all courses at Syracuse University. Students are expected to arrive on campus in time to attend the first meeting of all classes for which they are registered. Students who do not attend classes starting with the first scheduled meeting may be academically withdrawn as not making progress toward degree by failure to attend.

Academic Integrity Policy

As a pre-eminent and inclusive student-focused research institution, Syracuse University considers academic integrity at the forefront of learning, serving as a core value and guiding pillar of education. Syracuse University's Academic Integrity Policy provides students with the necessary guidelines to complete academic work with integrity throughout their studies. Students are required to uphold both course-specific and university-wide academic integrity expectations such as crediting your sources, doing your own work, communicating honestly, and supporting academic integrity. The full Syracuse University Academic Integrity Policy can be found by visiting class.syr.edu, selecting, "Academic Integrity," and "Expectations and Policy."

Upholding Academic Integrity includes the protection of faculty's intellectual property. Students should not upload, distribute, or share instructors' course materials, including presentations, assignments, exams, or other evaluative materials without permission. Using websites that charge fees or require uploading of course material (e.g., Chegg, Course Hero) to obtain exam solutions or assignments completed by others, which are then presented as your own violates academic integrity expectations in this course and may be classified as a Level 3 violation. All academic integrity expectations that apply to in-person assignments, quizzes, and exams also apply online.

Students found in violation of the policy are subject to grade sanctions determined by the course instructor and non-grade sanctions determined by the School or College where the course is offered. Students may not drop or withdraw from courses in which they face a suspected violation. Any established violation in this course may result in course failure regardless of violation level.

Use of Artificial Intelligence (AI) in Course

Based on the specific learning outcomes and assignments in this course, artificial intelligence is permitted to a limited extent on in-class labs, take-home assignments, and the final project. See each assignment's instructions for more information about what artificial intelligence tools are permitted and to what extent, as well as citation requirements. If no instructions are provided for a specific assignment, then no use of any artificial intelligence tool is permitted. Any AI use beyond that which is detailed in course assignments is explicitly prohibited except when documented permission is granted.

Conduct of Online Assessments

All academic integrity expectations that apply to in-person quizzes and exams also apply to online quizzes and exams. In this course, all work submitted for quizzes, assignments, and exams must be yours alone. Discussing quiz or exam questions with anyone during the quiz or exam period violates academic integrity expectations for this course.

Using websites that charge fees or require uploading of course material (e.g., Chegg, Course Hero) to obtain exam solutions or assignments completed by others and present the work as your own violates academic integrity expectations in this course and may be classified as a Level 3 violation, resulting in suspension or expulsion from Syracuse University.

Diversity and Disability

(ensuring that students are aware of their rights and responsibilities in a diverse, inclusive, accessible, bias-free campus community) can be found here.

Religious Observances Notification and Policy

Syracuse University's Religious Observances Policy recognizes the diversity of faiths represented in the campus community and protects the rights of students, faculty, and staff to observe religious holy days according to their traditions. Under the policy, students are given an opportunity to make up any examination, study, or work requirements that may be missed due to a religious observance, provided they notify their instructors no later than the academic drop deadline. For observances occurring before the drop deadline, notification is required at least two academic days in advance. Students may enter their observances in MySlice under Student Services/Enrollment/My Religious Observances/Add a Notification.

Disability-Related Accommodations

Syracuse University values diversity and inclusion; we are committed to a climate of mutual respect and full participation. There may be aspects of the instruction or design of this course that result in barriers to your inclusion and full participation in this course. I invite any student to contact me to discuss strategies and/or accommodations (academic adjustments) that may be essential to your success and to collaborate with the Center for Disability Resources (CDR) in this process.

If you would like to discuss disability-accommodations or register with CDR, please visit Center for Disability Resources. Please call (315) 443-4498 or email disabilityresources@syr.edu for more detailed information.

The CDR is responsible for coordinating disability-related academic accommodations and will work with the student to develop an access plan. Since academic accommodations may require early planning and generally are not provided retroactively, please contact CDR as soon as possible to begin this process.

Discrimination or Harassment

Federal and state law, and University policy prohibit discrimination and harassment based on sex or gender (including sexual harassment, sexual assault, domestic/dating violence, stalking, sexual exploitation, and retaliation). If a student has been harassed or assaulted, they can obtain confidential counseling support, 24-hours a day, 7 days a week, from the Sexual and Relationship Violence Response Team at the Counseling Center (315-443-8000, Barnes Center at The Arch, 150 Sims Drive, Syracuse, New York 13244). Incidents of sexual violence or harassment can be reported non-confidentially to the University's Title IX Officer (Sheila Johnson Willis, 315-443-0211, titleix@syr.edu, 005 Steele Hall). Reports to law enforcement can be made to the University's Department of Public Safety (315-443-2224, 005 Sims Hall), the Syracuse Police Department (511 South State Street, Syracuse, New York, 911 in case of emergency or 315-435-3016 to speak with the Abused Persons Unit), or the State Police (844-845-7269). I will seek to

keep information you share with me private to the greatest extent possible, but as a professor I have mandatory reporting responsibilities to share information regarding sexual misconduct, harassment, and crimes I learn about with the University's Title IX Officer to help make our campus a safer place for all.

Health and Wellness

Mental health and overall well-being are significant predictors of academic success. As such it is essential that during your college experience you develop the skills and resources effectively to navigate stress, anxiety, depression, and other mental health concerns. Please familiarize yourself with the range of resources the Barnes Center provides (https://ese.syr.edu/bewell/) and seek out support for mental health concerns as needed. Counseling services are available 24/7, 365 days, at 315-443-8000.

Course Evaluations

At the end of the term, the iSchool will ask you to share course feedback through EvaluationKIT [https://coursefeedback.syr.edu]. Log in to EvaluationKIT using your NetID and password. Please take the time to share your feedback about this course and your experience in it; all ratings and comments are completely anonymous. The iSchool carefully reviews your feedback. Our instructors use this feedback to fine tune course delivery and instruction; our professors of record use this feedback to fine tune course content and assignments. All feedback is factored into iSchool decisions about course, program, and instructor development.

Use of Blackboard

This class will use the Blackboard Learning Management to house the syllabus, course content, links to external course materials, assignments, quizzes, exams, feedback, and grades. Note when submitting materials that the University's Blackboard Learning Management System is on Eastern Time.

Information about Blackboard is available on Answers Blackboard; alternatively, you can contact Information Technology Services by sending an email to help@syr.edu, calling 315.443.2677, or in-person at the ITS Service Center, located at 1-227 CST in the Life Sciences Complex. Business hours for the Service Center can be found on the ITS Website at http://its.syr.edu/its_service_center/

Academic work

Academic work completed during a semester may be used by professors for educational purposes in courses during the semester. Students' registration and continued enrollment constitute consent for this purpose. Before using students' work for educational purposes in subsequent semesters, professors will either request students' permission in writing or render the work anonymous by removing all personal identification.